

Srikar Jakkena

Python Developer | AI & ML Engineer

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Karimnagar, Telangana, India

Professional Summary

Computer Science graduate specializing in AI & Machine Learning, with hands-on experience across **Python**, **Deep Learning**, and **Generative AI**. Skilled in building end-to-end ML pipelines with **TensorFlow** and **Scikit-learn**, and applying LLM-based tools – **LangChain**, **prompt engineering**, and **retrieval-augmented generation (RAG)** – to real-world problems. Combines a strong CV/DL foundation with growing Generative AI engineering skills to deliver production-ready, explainable AI systems.

Education

Kakatiya Institute of Technology & Sciences

Bachelor of Technology in Computer Science – AI & Machine Learning | CGPA: 8.14

Warangal, India

Nov 2022 – May 2026

Alphores Junior College

Intermediate Public Examination (IPE) | Percentage: 90.9%

Karimnagar, India

Jul 2020 – Mar 2022

Sri Kakatiya High School

Secondary School Certificate (SSC) | GPA: 10.0

Karimnagar, India

Jun 2019 – Mar 2020

Projects

RAG-Based Document Q&A Chatbot | Python, LangChain, Gemini/OpenAI API, Streamlit | [\[View\]](#) 2026

- Built a **retrieval-augmented generation (RAG)** chatbot enabling users to upload PDF documents and ask natural-language questions, with responses grounded in retrieved source content.
- Implemented a document ingestion pipeline with **text chunking**, **embedding generation**, and vector storage/retrieval using **ChromaDB** to reduce hallucinated responses.
- Integrated **LangChain** for prompt orchestration and multi-step retrieval-generation flow, deployed with a **Streamlit** UI.

Hybrid Brain Tumor Detection & Classification System | Python, TensorFlow, Keras, Scikit-learn | [\[View\]](#) 2025

- Designed a hybrid AI pipeline combining **U-Net** segmentation for tumor boundary detection and **MobileNetV2** for deep feature extraction on **3,000+ MRI images** and BraTS Dataset.
- Implemented **SVM** and **Random Forest** classifiers to predict tumor types (Glioma, Meningioma, Pituitary, No Tumor) with **96% classification accuracy** on benchmark MRI datasets.
- Integrated **Grad-CAM** for explainable AI visualization and built an automated workflow for MRI preprocessing, tumor localization, classification, and result reporting.

Real-Time Sign Language Recognition System | Python, TensorFlow, CNN, OpenCV | [\[View\]](#) 2024

- Built an end-to-end **CNN-based** gesture recognition system, translating American Sign Language gestures to text with **92% accuracy** in real-time.
- Designed a custom data preprocessing pipeline with frame extraction, normalization, and augmentation to improve model generalization across diverse lighting conditions and hand positions.

Experience

Front-End Web Developer Intern

Jul 2024 – Jan 2025

Deify Infocomm Solutions Private Limited

Hyderabad, India

- Architected and deployed an IP-based real-time video conferencing platform, achieving a **30% increase in user engagement** during beta testing with **200+ users**.
- Developed responsive UI components for a real-time communication platform using **HTML**, **CSS**, **JavaScript**, and **jQuery**.
- Reduced page load time by optimizing CSS and JavaScript bundles, improving performance across **5+ responsive UI components**.

Technical Skills

Programming Languages: Python, Java, SQL, HTML5, CSS3, JavaScript

Generative AI: LangChain, Prompt Engineering, RAG (Retrieval-Augmented Generation), LLM APIs (OpenAI, Gemini, Anthropic Claude, Groq, Hugging Face), Ollama (local LLM deployment)

Agentic AI: LangGraph, AI Agents, Tool-Calling, Multi-Step Reasoning

Libraries & Frameworks: TensorFlow, Keras, Scikit-learn, OpenCV, Pandas, NumPy, Matplotlib

Architectures & Concepts: Deep Learning, CNNs, U-Net, MobileNetV2, NLP, Model Interpretability (Grad-CAM)

Databases: MySQL, ChromaDB (Vector DB)

Tools & Cloud: AWS, GitHub, VS Code, Jupyter

Certifications

Front-End Web Development -- Deify Infocomm Solutions Private Limited, 2025

AI Upskilling Certificate: Hands-On Development from Model to App – Qualcomm Academy